



< Previous







Next >

10.3.4 The implicit Q theorem

 Bookmark this page

Pursue a verified certificate



Verified Certificate

THE UNIVERSITY of TEXAS SYSTEM

This is to certify that

Sandipan Dey

successfully completed and received a passing grade in

UT.ALA:Advanced Linear Algebra: Foundations to Frontiers

a course of study offered by UTAustinX, an online learning initiative of University of Texas System.

**Maurilio Pugliesi**
Professor
UniversityX

**Justine Doe, Ph.D.**
Professor
UniversityX

Verified Certificate

Valid Certificate ID

Issued Month / Year

12345678901234567890

- Unlimited access to all course materials
- Shareable [verified certificate](#)
- AI learning assistant and translations

Upgrade for \$99

Video



▶

6:55 / 6:55

▶ 2.0x

🔊

🔌

CC

🗣️

plus $\beta_1 u$
times q_1 . Okay? So
these two must be
equal. Now A is
known, q_0 is known,
 q_0 is
known here. Hmm,
we know that q_0 is
orthogonal to q_1 . So
just like when we
came up with the
algorithm for
computing the Gram-
Schmidt process, we
say, "Oh, we
can then compute
 β_0 by saying
let's look at $q_0^T A q_0$."
That
must be equal to
 β_0 .
So at this point we've

Video

📄 [Download video file](#)

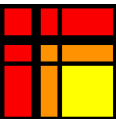
Transcripts

📄 [Download SubRip \(.srt\) file](#)

📄 [Download Text \(.txt\) file](#)

Reading assignment

1/1 point (graded)



Read Unit 10.3.4 of the notes. [\[LINK\]](#)

☒ done



Submit

✓ Correct (1/1 point)

Discussion

Hide Discussion

Topic: Week 10 / 10.3.4

Add a Post

Show all posts ▼by recent activity ▼

There are no posts in this topic yet.

✕

Homework 10.3.4.1

1/1 point (graded)

Give all the details of the proof of Theorem 10.3.4.2 in [the notes](#).

☒ done



Solution:

For a complete solution, see [the notes](#).

Submit

Answers are displayed within the problem

Video

- if we actually tried to form with you then we would

have to take our matrix G_0 and hit it from the right with the next Givens rotation. But remember that Givens rotation when applied from the right only affects the second and third column. Importantly it leaves the first column alone. So you get a bunch of non-zeros here. As a matter of fact, you get a non-zero here, here, and here, typically. Okay? Then you go to the

Video

< Previous

Next >



edX

- [About](#)
- [Affiliates](#)
- [edX for Business](#)
- [Open edX](#)

Legal

- [Terms of Service & Honor Code](#)
- [Privacy Policy](#)
- [Accessibility Policy](#)
- [Trademark Policy](#)
- [Sitemap](#)
- [Cookie Policy](#)
- [Your Privacy Choices](#)

Connect

- [Idea Hub](#)
- [Contact Us](#)
- [Help Center](#)
- [Security](#)
- [Media Kit](#)

